ARDUINO

Lec1

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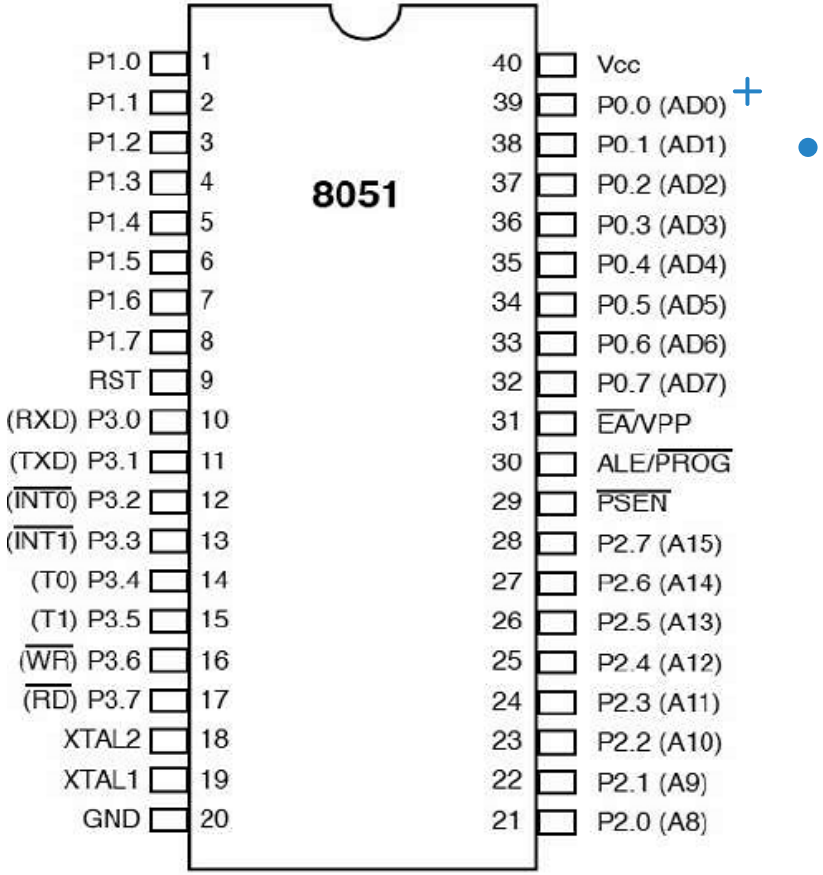
❑ Introduction

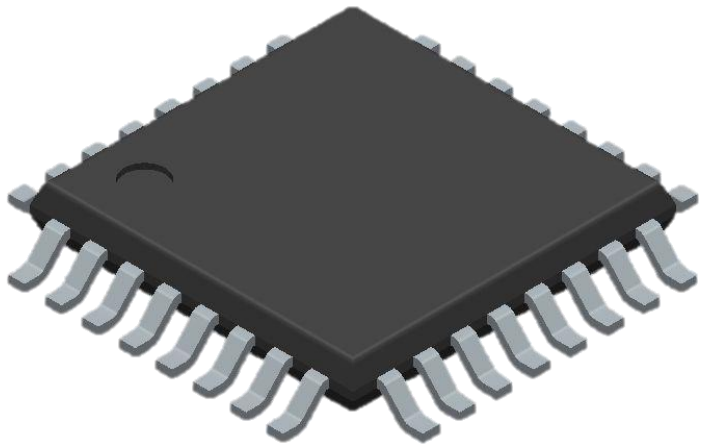
- In this course we will learn about the Arduino which consists of micro controller which is a simple part of the embedded system .
- The micro controller's structure is in the right of page.
- The micro controller consists of pins for VCC and GND and GPIO .

  GPIO means (General Purpose Input Output)
- The micro controller has a clock pin , so that we want a clock generator (crystal oscillator) .



Micro Controller

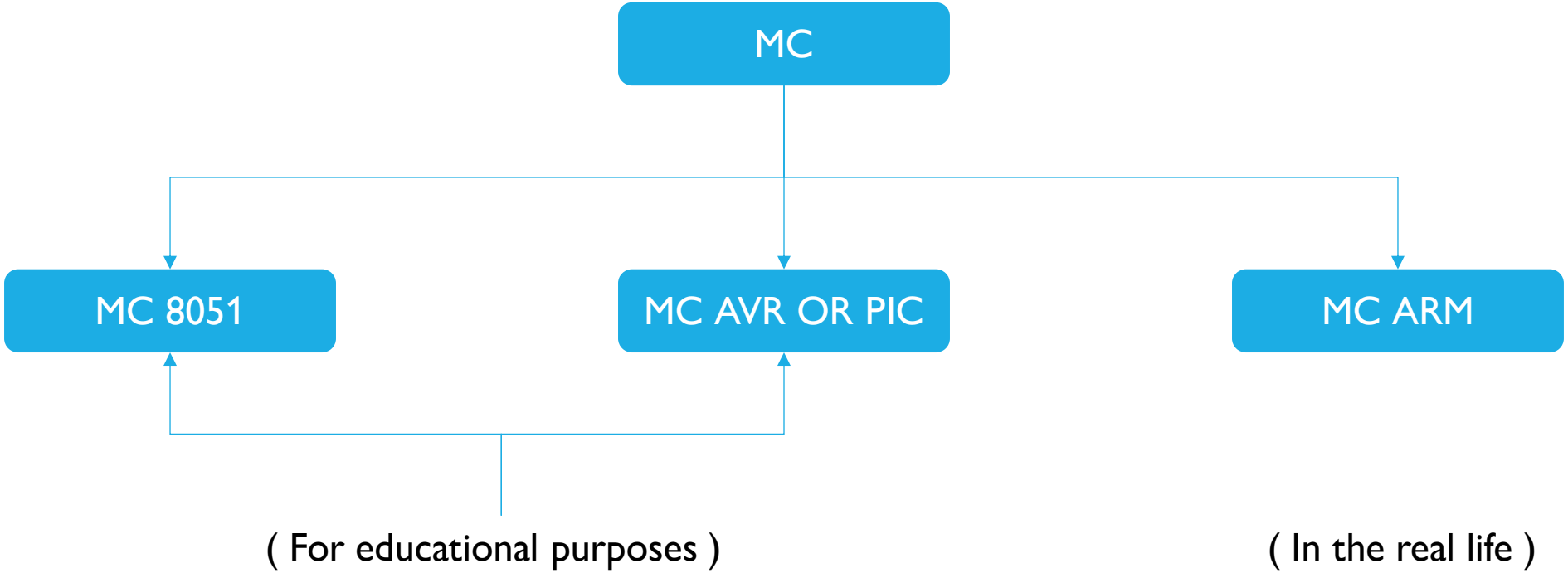
Crystal Oscillator

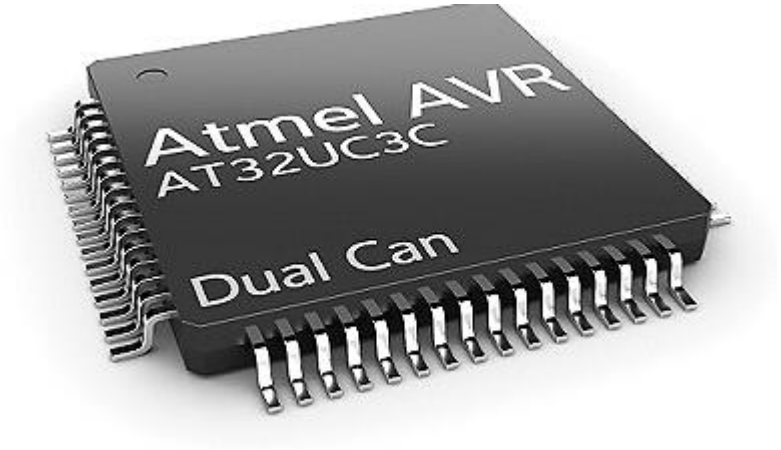


Micro Controller

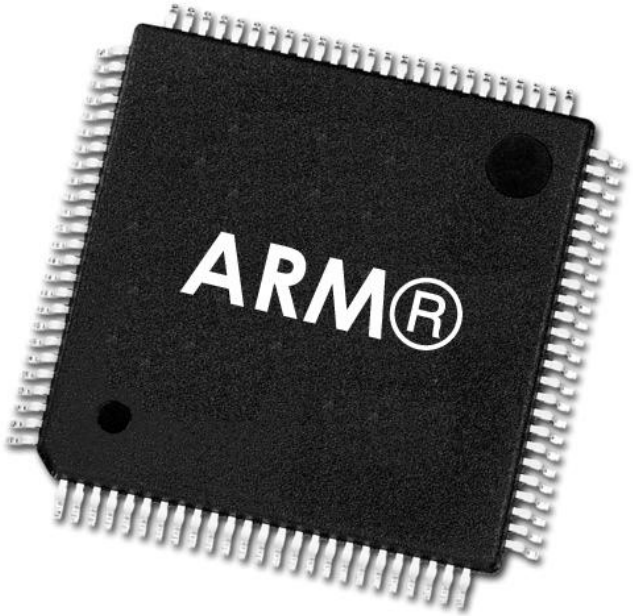
MICRO CONTROLLERS

❑ Types of micro controllers





MC AVR



MC ARM

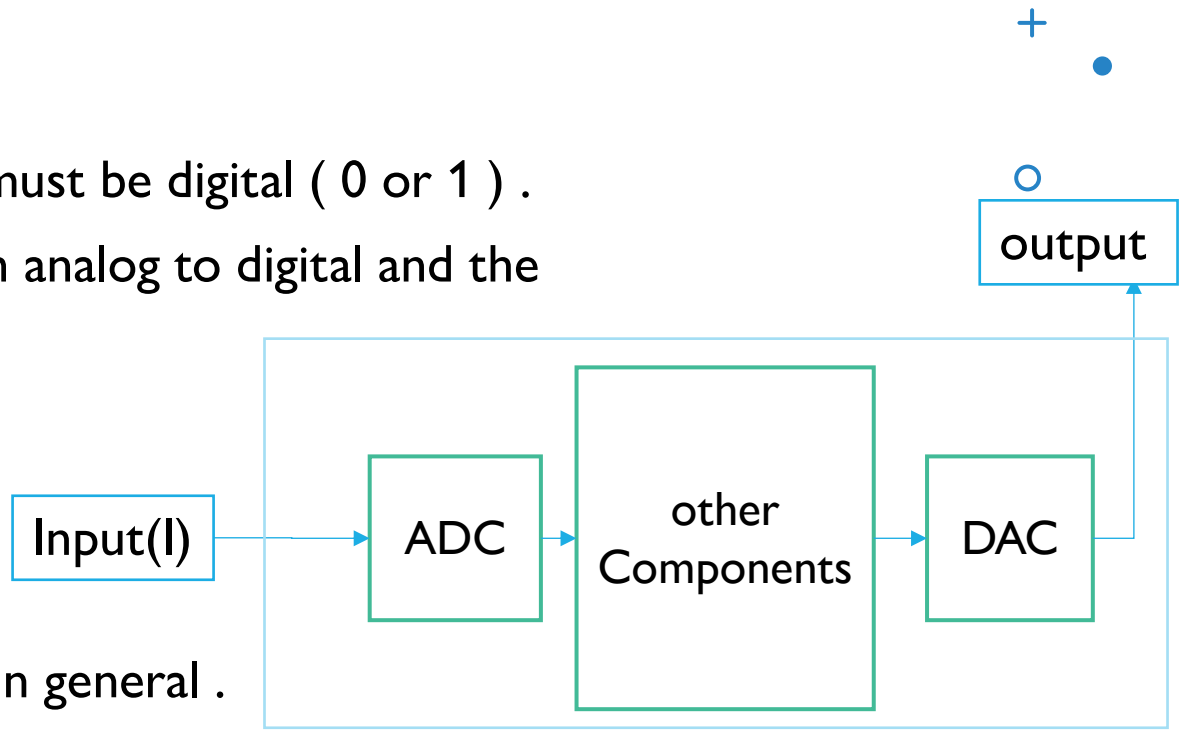
MC 8051

✓ Notes :

- The inputs and the outputs of the micro controller must be digital (0 or 1) .
- There is a signal converter to convert the signal form analog to digital and the reverse .

    1. ADC → analog to digital converter .
    2. DAC → digital to analog converter .

- Some micro controllers has a ADC to use as inputs in general .
- The input (I) can be digital or analog .



(General Block Micro Controller)

❑ How to program the MC ??

- Any micro controller can be programmed by C or assembly language .
- Any micro controller is a small computer (has cpu and ram and rom and inputs and outputs)

Hint :
There are 2 types of programming language :

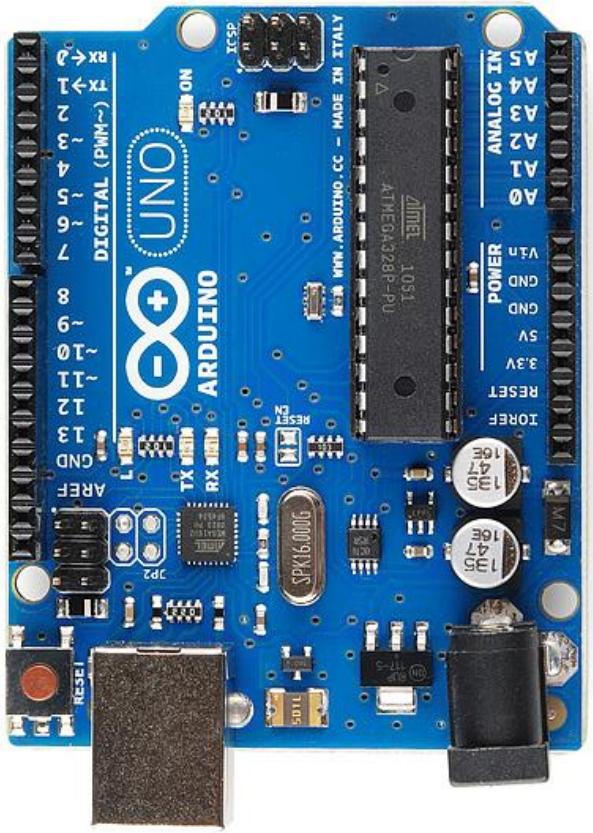
1. Low level languages : such as Machine Language and Assembly Language
2. High level languages such as : Python and C and C++

Any CPU converts the high level language to assembly then to machine language

ARDUINO

❑ Arduino

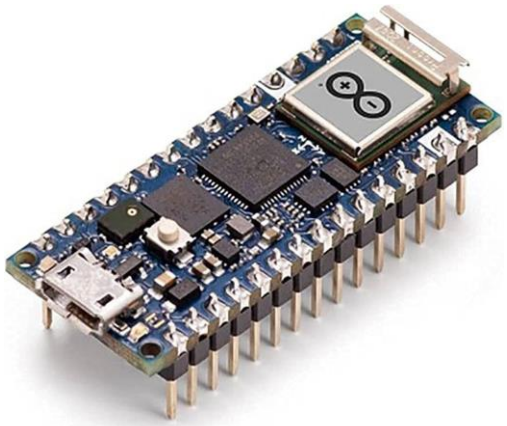
- Any Arduino consists of a micro controller with some features .
- Arduino has a input for power and an input to connect with the computer to burn the codes .
- there is reset button in the side to reset the Arduino .
- There is a led to show the if the Arduino burns the code or not .
- The micro controller here in the Arduino is ATMEGA 32 is a type of AVR micro controllers .



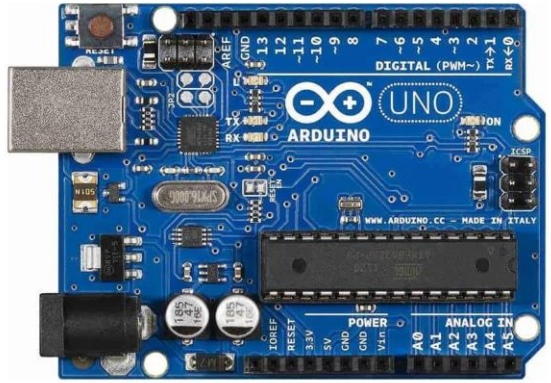
(ARDUINO UNO)

❑ Types of Arduino ??

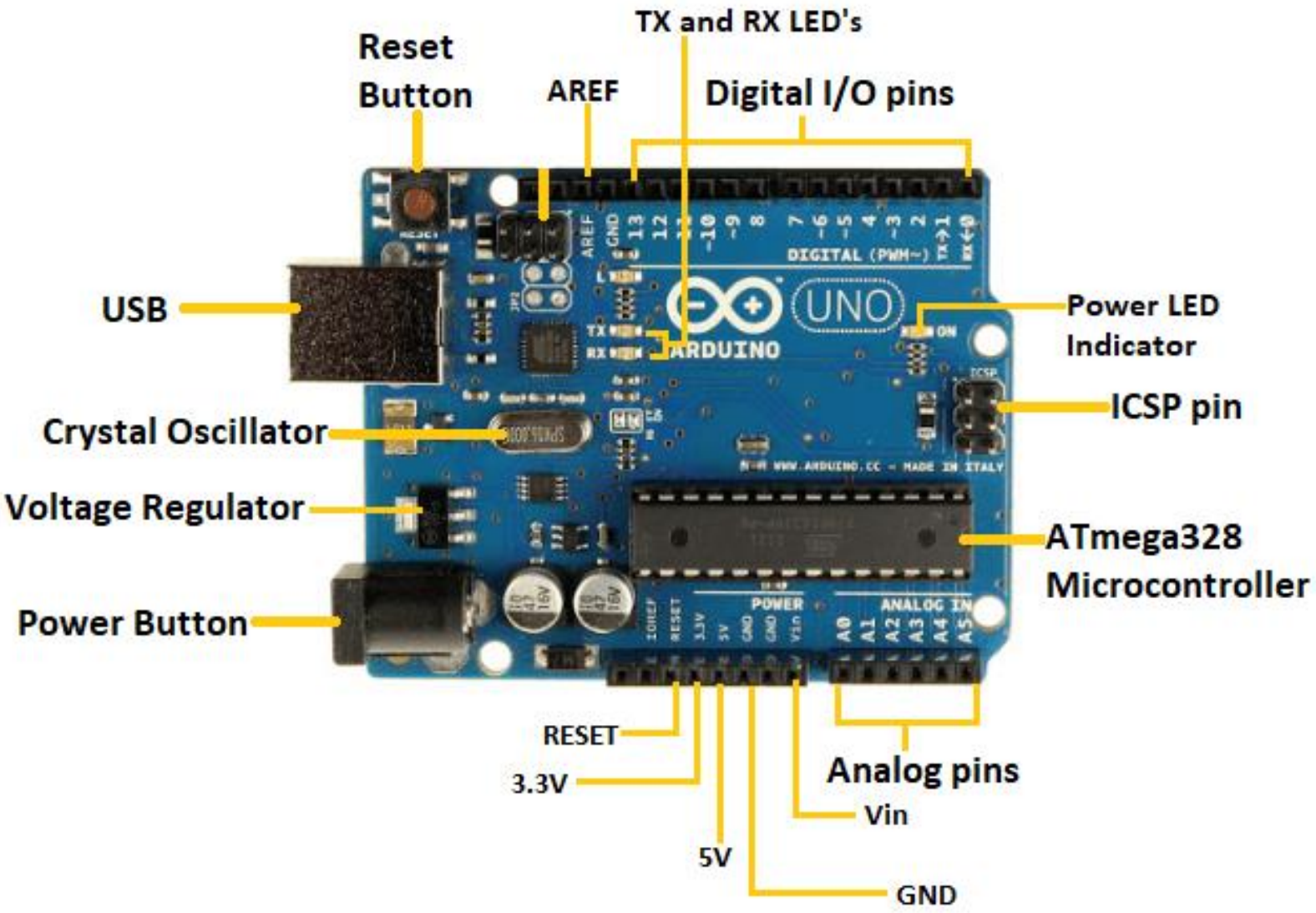
Mega Arduino



Nano Arduino



Uno Arduino



(Construction Of The Arduino)

HOW TO CODE THE ARDUINO ?

- We will program the Uno Arduino using C++ language .
- We will use ARDUINO IDE to code .

❑ How the code will be run ??

- As we know in C++ the main function only will be on automatically .
- In Arduino code we need the Arduino the repeat an operation all time .
- so that the code should be the following :

```
Void main (){
        -----
        initialization ;
        -----
        while true :

                the code of Arduino;
}
```

- Now we have known the code of main only will be on automatically .
- So that to code the Arduino we need to write 2 parts m one for initialization and the other for the main function in the loop , so that the ARDUINO IDE program will help us by dividing the program into 2 functions and in the back ground the program calls the 2 functions in the main .
- In the ARDUINO IDE →

```
Void setup( ){
        the initialization
}
Void loop( ){
        the main code of Arduino
}
```

In the back →

```
Void main () {

        setup()
        while ( 1 ):
                loop( )

}
```

(this is the code in the back)

✓ Ex :-

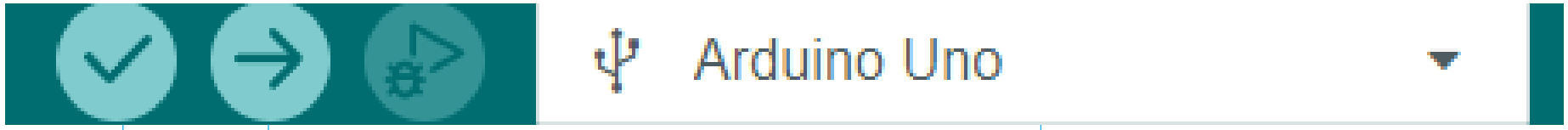
```
void setup() {
  // put your setup code here, to run once:
  pinMode(13,OUTPUT);

}

void loop() {
  // put your main code here, to run repeatedly:
    // the led will be on for 5000 ms (5 seconds)
    digitalWrite(13,HIGH);
    delay(5000);

    // the led will be off for 1000 ms (1 seconds)
    digitalWrite(13,LOW);
    delay(1000);

}
```

pinMode(no. of pin , Mode (in or ou))
(To define the pin as in or output)

digitalWrite(no. of pin , (High or Low))
(to set a low or high value on the pin)

delay(the period of sleeping in milli seconds)

✓ Some notes in use of program :



To burn the code on Arduino

Choose your Arduino to burn on

To verify the code

THANK YOU

FOR PAYING ATTENTION